

CURATED BEGINNER WORKSHEET

Chapter 13: Fundamentals of Algebraic Expressions & Polynomials

Name: _____

Date: _____

Max Marks: 40

SECTION A: Multiple Choice Questions (8 Marks)

Choose the correct option for each of the following questions. Each question carries 1 mark.

Q1. Which of the following represents a **variable** (or literal) in algebra? 4

- (a) 5
- (b) -2
- (c) x
- (d) $\frac{3}{7}$

Q2. What is the **numerical coefficient** in the term $-7x^2y$? 4

- (a) -7
- (b) 7
- (c) x^2
- (d) y

Q3. The algebraic expression $3a + 5b - 2c$ consists of three terms. It is classified as a: 4

- (a) Monomial
- (b) Binomial
- (c) Trinomial
- (d) Constant

Q4. Which of the following pairs contains **like terms**? 4

- (a) $3x, 3y$
- (b) $-5ab, 2ba$
- (c) $4a^2b, 4ab^2$
- (d) $x^2, 2x$

Q5. What is the **degree** of the polynomial $4x^3 - 2x^2 + 7x - 5$? 4

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Q6. The verbal statement "5 subtracted from 3 times a number n " is represented algebraically as: 4

- (a) $5 - 3n$
- (b) $3n - 5$
- (c) $3(n - 5)$
- (d) $5n - 3$

Q7. Simplifying the sum of like terms $4x + 9x$ yields: 4

- (a) $13x^2$
- (b) $36x$
- (c) $13x$
- (d) $5x$

Q8. If $p = 3$, what is the numerical value of the expression $2p^2 - 4$? 4

- (a) 8
- (b) 14
- (c) 2
- (d) 32

SECTION B: Concept Identification & Short Answer (8 Marks)

Answer the following questions clearly.

Q9. Classification [2 Marks]: Classify each of the following expressions as a *Monomial*, *Binomial*, or *Trinomial*:

- | | |
|----------------|--------------------------------------|
| (i) $12x^2y$ | (iii) $p^2 + 4p - 12$ |
| (ii) $5a - 8b$ | (iv) $7x^2 - 3x^2$ (Simplify first!) |

Q10. Coefficients Breakdown [2 Marks]: Write down the **numerical coefficient** and the **literal coefficient** for each monomial:

- | | |
|--------------|--------------------------|
| (i) $-9pq^2$ | (ii) $\frac{4}{7}a^2b^3$ |
|--------------|--------------------------|

Q11. Identifying Like Terms [2 Marks]: Group the like terms together from the given set:

$$\{-3xy^2, \quad 5x^2y, \quad 7xy^2, \quad -2x^2y, \quad 8xy\}$$

Q12. Polynomial Check & Degree [2 Marks]: State whether each expression is a polynomial. If yes, state its degree:

- | | |
|----------------------|-------------------------------|
| (i) $6x^3 - 3x + 10$ | (ii) $4y^2 + \frac{5}{y} - 2$ |
|----------------------|-------------------------------|

SECTION C: Generating Algebraic Expressions (8 Marks)

Translate each word statement into a mathematical algebraic expression.

Q13. The product of a and b added to their sum.

Q14. Thrice x increased by 9.

Q15. One-fourth of y subtracted from 15.

Q16. The total cost of x notebooks at 25 each and y pens at 10 each.

SECTION D: Operations & Problem-Based Questions (16 Marks)*Show full working for each problem.*

Q17. Addition of Polynomials [4 Marks]: Add the following algebraic expressions:

$$(4x^2 - 6x + 5) \quad \text{and} \quad (3x^2 + 9x - 8)$$

Q18. Subtraction of Polynomials [4 Marks]: Subtract $(3a - 5b + 4c)$ from $(8a + 2b - c)$.

Q19. Geometry Application [4 Marks]: The three side lengths of a triangle are $(2x + y)$, $(3x - 2y)$, and $(x + 4y)$. Find an algebraic expression for the **perimeter** of the triangle in simplified form.

Q20. Substitution & Value Evaluation [4 Marks]: If $m = 2$ and $n = -3$, evaluate the numerical value of the expression:

$$3m^2 - 2mn + 5n$$

COMPLETE ANSWER KEY & SOLUTIONS

Section A: MCQs

- Q1. (c) x** — A symbol that represents a variable value is a literal/variable. Numbers $(5, -2, \frac{3}{7})$ are constants.
- Q2. (a) -7** — The constant numerical multiplier attached to the variables is -7 .
- Q3. (c) Trinomial** — The expression consists of 3 distinct non-like terms $(3a, 5b, -2c)$.
- Q4. (b) $-5ab, 2ba$** — Both terms have identical literal factors ($ab = ba$).
- Q5. (c) 3** — The highest power of the variable x in the expression is 3.
- Q6. (b) $3n - 5$** — "3 times n " is $3n$, and subtracting 5 gives $3n - 5$.
- Q7. (c) $13x$** — Combining numerical coefficients: $(4 + 9)x = 13x$.
- Q8. (b) 14** — Substituting $p = 3$: $2(3)^2 - 4 = 2(9) - 4 = 18 - 4 = 14$.

Section B: Short Answer Solutions

- Q9.** (i) Monomial (1 term), (ii) Binomial (2 terms), (iii) Trinomial (3 terms), (iv) $7x^2 - 3x^2 = 4x^2 \rightarrow$ Monomial (1 term).
- Q10.** (i) Numerical coefficient = -9 , Literal coefficient = pq^2
 (ii) Numerical coefficient = $\frac{4}{7}$, Literal coefficient = a^2b^3
- Q11.** Group 1 (like terms in xy^2): $-3xy^2$ and $7xy^2$
 Group 2 (like terms in x^2y): $5x^2y$ and $-2x^2y$
 Unpaired term: $8xy$
- Q12.** (i) **Yes**, it is a polynomial in x with non-negative integer powers. Degree = 3.
 (ii) **No**, because $\frac{5}{y} = 5y^{-1}$ contains a negative integer exponent.

Section C: Generating Expressions

- Q13.** $(a + b) + ab$
- Q14.** $3x + 9$
- Q15.** $15 - \frac{y}{4}$ or $15 - \frac{1}{4}y$
- Q16.** Total cost = $(25x + 10y)$

Section D: Problem-Based Solutions

Q17. Addition:

$$(4x^2 - 6x + 5) + (3x^2 + 9x - 8) = (4x^2 + 3x^2) + (-6x + 9x) + (5 - 8) = \mathbf{7x^2 + 3x - 3}$$

Q18. Subtraction:

$$(8a + 2b - c) - (3a - 5b + 4c) = 8a + 2b - c - 3a + 5b - 4c = (8a - 3a) + (2b + 5b) + (-c - 4c) = \mathbf{5a + 7b - 5c}$$

Q19. Perimeter of Triangle:

$$\text{Perimeter} = \text{Side}_1 + \text{Side}_2 + \text{Side}_3 = (2x + y) + (3x - 2y) + (x + 4y)$$

$$\text{Grouping like terms : } (2x + 3x + x) + (y - 2y + 4y) = \mathbf{6x + 3y}$$

Q20. Evaluation:

$$\text{Given } m = 2, n = -3 :$$

$$3m^2 - 2mn + 5n = 3(2)^2 - 2(2)(-3) + 5(-3) = 3(4) - (-12) - 15 = 12 + 12 - 15 = \mathbf{9}$$